Kalshi Tmax Market Strategy — Research Log

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2026-06-14

1 Data and Market Structure

1.1 Overview

This project evaluates trading strategies for Kalshi same-day maximum temperature (Tmax) prediction markets across 12 US cities. Each trading day, Kalshi offers a set of mutually exclusive 2°F-wide temperature buckets covering the plausible Tmax range. One bucket resolves to $1 (YES wins); all others resolve to $0. Settlement uses the official NWS Daily Climate Report (CLI) for each city's ASOS station. The dataset contains 5-minute market snapshots from March 30 to June 2, 2026, restricted to the post-10:00 AM local time trading window. Table 1a summarises the city-level inventory.

1.2 Evaluation Design

The dataset is split along both time and location axes (Figure 1). Nine train cities are used for model development; three holdout cities (Denver, Miami, Minneapolis) are reserved for generalisation testing and are never used for parameter tuning. The time axis assigns the most recent 15 calendar dates to an out-of-sample (OOS) partition (time_holdout), with earlier dates forming the in-sample partition (threshold_opt). A final true_holdout partition (holdout cities × OOS dates) is never inspected until final reporting.

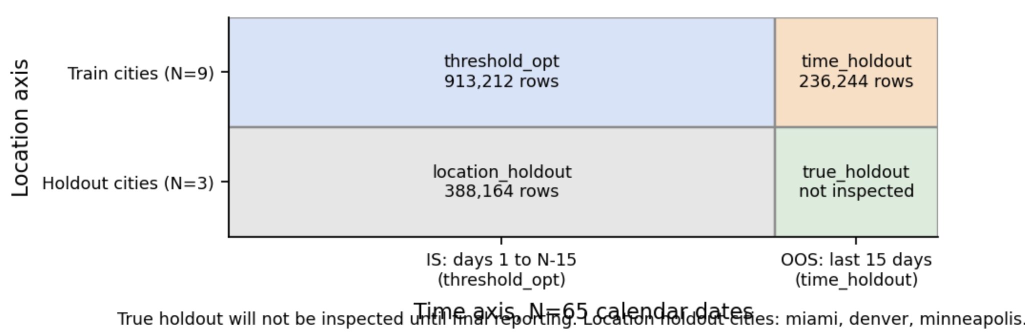

Figure 1

1.3 Fee Model

All profit and loss figures are reported net of Kalshi fees. The taker fee in cents is $\lceil 0.07 \cdot C \cdot P \cdot (1-P) \rceil$ and the maker fee is $\lceil 0.0175 \cdot C \cdot P \cdot (1-P) \rceil$, where C is the number of contracts and P is the YES mid price. The ceiling function creates a 1-cent minimum per trade; for positions under approximately 57 contracts, the

Table 1

(a) City inventory with N days, N snapshots, and date range.

City	N days	N snapshots	First date	Last date
austin	56	16539	2026-03-30	2026-06-14
chicago_midway	68	20202	2026-03-30	2026-06-14
denver†	61	17918	2026-03-30	2026-06-02
houston	72	21102	2026-03-30	2026-06-14
los_angeles	72	21049	2026-03-30	2026-06-14
miami†	55	16099	2026-03-30	2026-05-30
minneapolis†	64	18601	2026-03-30	2026-06-02
new_york_city	71	20870	2026-03-30	2026-06-14
oklahoma_city	73	21470	2026-03-30	2026-06-14
philadelphia	62	18490	2026-03-30	2026-06-14
phoenix	74	21767	2026-03-30	2026-06-14
san_francisco	73	21434	2026-03-30	2026-06-14

† Held-out city: not used for parameter tuning.

fee is pinned at 1 cent regardless of size. Figure 2 shows the fee schedule for $C = 100$ contracts. A no-trade guardrail skips any bucket where the estimated edge $(p - c)$ is less than twice the per-contract fee, and a price floor excludes buckets priced below \$0.15.

2 Market Baselines and Efficiency

2.1 Baseline Definitions

Seven market baselines test whether simple rules extract profit from the Tmax bucket structure. Entry timing uses a snapshot-stability rule: the first post-10:00 AM snapshot at which the modal bucket (highest YES mid price) has been unchanged for $k = 1$ consecutive 5-minute interval, which in practice fires immediately at market open. The implied-favorite baseline enters long on the modal bucket at this snapshot. The distribution-copy baseline allocates capital across all buckets proportional to fee-adjusted price, serving as a theoretical zero-edge benchmark. The sell-longshots baseline shorts any bucket crossing below \$0.10 after the stability snapshot, motivated by the documented favorite–longshot bias. The make-the-market baseline quotes one cent inside the modal bucket spread until a maker fill arrives. Three signal-threshold baselines enter when mode probability exceeds $t^* = 0.88$ (mode-prob), Shannon entropy falls below $H^* = 0.45$ (entropy), or mode probability momentum exceeds $d^* = 0.02$ over $w^* = 16$ snapshots (momentum). Threshold values were frozen on the in-sample partition.

2.2 Intraday Market Dynamics

Figure 3 shows the average post-10 AM intraday mode probability across all train-city trading days, illustrating how the market concentrates toward a dominant bucket as the day progresses.

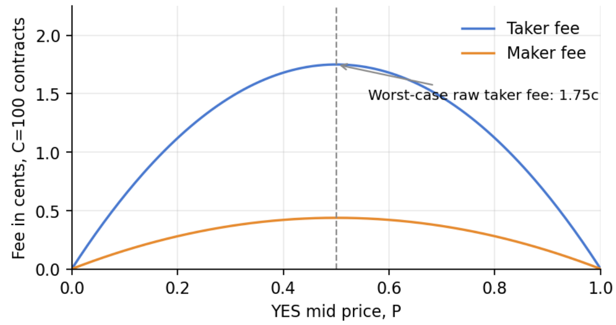

Figure 2: Kalshi fee schedule for C=100 contracts. Fees peak at P=0.50 and are substantially lower near the tails.

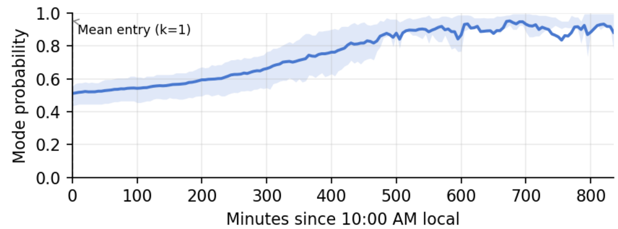

Figure 3: Average intraday mode probability across all train-city trading days. The shaded band shows the 25th-75th percentile. The dashed line marks the mean snapshot at which the stability rule fires.

Table 2

(a) IS and OOS statistics for all seven market baselines. IS uses threshold_opt (416 day-equiv); OOS uses time_holdout (108 day-equiv). No re-optimisation occurred between partitions.

Baseline	IS N	OOS N	IS SR	OOS SR	OOS CI	PSR(0)	Sortino	Max DD
implied favorite	412	107	-0.35	1.16	[-1.84, 4.16]	0.77	6.39	-3.79
distribution copy	412	107	-9.71	-6.05	[-9.15, -2.95]	0.00	-5.91	-4.58
sell longshots	388	97	-0.28	-0.60	[-3.60, 2.39]	0.34	-0.63	-3.54
make the market	411	107	-0.45	1.45	[-1.55, 4.45]	0.82	8.13	-3.79
mode prob threshold	170	50	-0.83	0.10	[-2.89, 3.09]	0.53	0.77	-1.40
entropy threshold	139	44	0.42	-0.49	[-3.48, 2.51]	0.37	-1.96	-1.81
momentum threshold	400	104	-0.52	1.36	[-1.63, 4.36]	0.81	6.98	-4.07

2.3 Baseline Results

Table 2a reports IS and OOS statistics for all seven baselines. No baseline achieves a positive deflated Sharpe or meets the minimum track record length (MinTRL) required for statistical significance at the 5% level. The Spearman rank correlation between IS and OOS Sharpe is 0.036, indicating that IS rankings provide essentially no information about OOS performance. Three baselines show positive OOS point estimates (make-the-market 1.45, momentum 1.36, implied-favorite 1.16), but all 95% confidence intervals include zero. Furthermore, 138% of combined OOS PnL is concentrated in the top three trading days, suggesting these positive results are driven by a small number of favorable days rather than a persistent signal.

2.4 IS versus OOS Comparison

Figure 4 compares IS and OOS Sharpe across baselines; the pronounced rank inversion and wide OOS confidence intervals underscore the preliminary nature of any performance claim on 108 day-equivalents.

3 Forecast Models

3.1 Architecture

Track-B is a three-learner ensemble that predicts next-day maximum temperature in degrees Fahrenheit for each of the nine train cities. The ensemble combines three regressors chosen for complementary inductive biases. Ridge regression provides a regularised linear baseline, with the penalty parameter α selected from

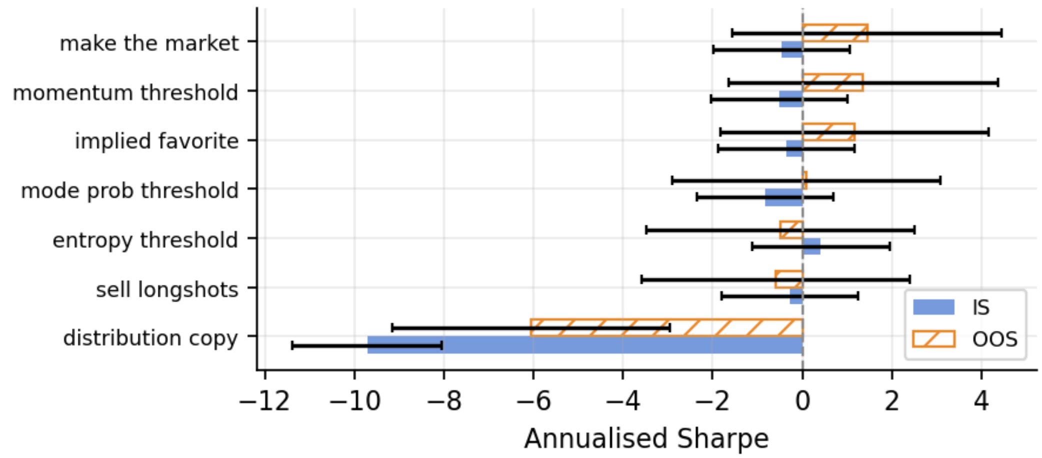

Figure 4

$\{0.01, 0.1, 1.0, 10.0, 100.0\}$ by minimising validation MAE. Huber regression ($\epsilon = 1.35$) fits a linear model robust to outliers, downweighting days where the residual exceeds 1.35 standard deviations; this guards against anomalous Tmax events (e.g. cold fronts, record heat) that would distort ordinary least-squares. LightGBM captures nonlinear feature interactions that the linear learners miss, trained with 500 estimators, maximum depth 4, learning rate 0.05, and early stopping on the validation set (patience 30 rounds). Both Ridge and Huber are fit on standardised features via scikit-learn pipelines.

The final prediction is the unweighted mean of the three base learners, rounded to the nearest integer degree Fahrenheit. Ensemble weights are not tuned: equal weighting avoids overfitting the combination to the validation set, which is a particular concern given that the validation window (2025) spans only one year. Models are trained on 2021–2024 data, validated on 2025, and tested on 2026 onward. The training and validation sets are strictly non-overlapping, and no test-period data is used at any stage of model fitting or selection.

The model ingests 25 candidate features across five groups: (1) nine ASOS morning observations (temperature, dewpoint, humidity, pressure, wind, cloud cover as of 10:00 AM), (2) nine calendar and temperature-lag features (day-of-year harmonics, lagged Tmax from 1 to 7 days, rolling means), (3) NWS MOS Tmax forecast issued the previous evening, (4) three GFS afternoon forecast fields (2-m temperature, dewpoint, cloud cover from the 00Z cycle), and (5) best-available NWP Tmax from ECMWF IFS (priority) or GFS seamless. Per-city feature selection on the training set drops columns with more than 20% missingness or absolute Pearson correlation below 0.05 with Tmax, retaining a mean of 22 features per city. A leakage audit confirmed that all GFS features use forecast hours ($fxx > 0$) available before the 10:00 AM trading cutoff.

3.2 Bucket Probability Conversion

The integer-valued ensemble prediction $\hat{T}$ is converted to per-bucket probabilities via a Gaussian CDF assumption. For a RANGE bucket $[a, b]$: $P_k = \Phi((b - \hat{T})/\sigma) - \Phi((a - \hat{T})/\sigma)$. For LESS_THAN and GREATER_THAN tail buckets, the corresponding one-sided CDF applies. The scale parameter σ is back-solved from the test-set ± 1°F hit rate h using $h = 2\Phi(1/\sigma) - 1$. Cities with higher hit rates produce smaller σ and therefore more peaked (confident) bucket distributions. Table 3a reports the back-solved σ for each city; values range from 1.02°F (Phoenix, tightest) to 2.52°F (Philadelphia, widest), directly determining how aggressively the model concentrates probability on its predicted bucket and therefore how large the implied edge is when the model disagrees with the market.

Table 3

(a) Track-B per-city training summary (train < 2025, val = 2025, test ≥ 2026).

City	N train	N val	N test	N feat	Val MAE	Test MAE	Test ±1°F	Sigma
austin	655	342	146	22	1.76	2.21	36.3%	2.12
chicago midway	714	361	154	23	1.69	1.87	53.2%	1.38
houston	669	338	153	22	1.30	1.42	64.7%	1.08
los angeles	699	360	151	23	1.26	1.54	54.3%	1.34
new york city	713	360	156	22	1.43	1.76	54.5%	1.34
oklahoma city	716	353	154	22	1.86	2.57	42.9%	1.77
philadelphia	714	359	149	23	1.67	2.33	30.9%	2.52
phoenix	653	344	138	20	1.40	1.29	67.4%	1.02
san francisco	709	359	146	22	1.93	2.00	40.4%	1.89

Table 4

(a) Track-B versus aligned Track-A/Track-J and raw NWS comparison on the 2026+ test window. Austin baseline is Track-J.

City	Track-A MAE	Track-B MAE	Improvement	Track-A ±1°F	Track-B ±1°F	Raw NWS MAE
austin	1.84	2.21	-0.37	46.4%	36.3%	6.27
chicago midway	2.85	1.87	0.98	30.7%	53.2%	8.36
houston	2.79	1.42	1.36	32.9%	64.7%	4.45
los angeles	1.80	1.54	0.25	54.4%	54.3%	3.28
new york city	3.65	1.76	1.89	25.3%	54.5%	6.59
oklahoma city	4.39	2.57	1.82	20.7%	42.9%	7.62
philadelphia	3.94	2.33	1.61	15.9%	30.9%	6.65
phoenix	1.82	1.29	0.54	55.0%	67.4%	3.47
san francisco	2.49	2.00	0.49	30.9%	40.4%	3.45

3.3 Per-City Forecast Accuracy

Table 3a reports Track-B accuracy on the 2026 test window. Phoenix achieves the best performance (MAE 1.29°F, 67.4% ±1°F hit rate, $\sigma = 1.02$°F), followed by Houston (1.42°F, 64.7%, $\sigma = 1.08$°F). Five cities exceed 50% hit rate. Philadelphia is the weakest (30.9%, $\sigma = 2.52$°F). Table 4a compares Track-B against aligned Track-A baselines (scored on the same 2026 dates) and raw NWS MOS forecasts. Track-B improves over Track-A for eight of nine cities, with the largest gains at New York City (1.89°F MAE reduction), Oklahoma City (1.82°F), and Philadelphia (1.61°F). Austin is the sole exception: the earlier Track-J model (which uses a richer 6-learner architecture with HRRR/RAP features) retains a 0.37°F advantage over the standardised Track-B pipeline.

3.4 Feature Importance

Figure 5 shows LightGBM split importances averaged across all nine cities. The best-available NWP forecast (nwp_tmax_best_f) and GFS afternoon temperature (gfs_t2m_afternoon) dominate, confirming that Track-B is primarily a bias-corrected NWP forecast. The 10:00 AM surface temperature (temp_10am) ranks third, providing same-morning ground truth that the NWP models lack. No leakage features (resolved, settled, or bucket fields) appear in the retained sets. Figure 6 shows the per-city MAE comparison across Track-A/Track-J, Track-B, and raw NWS, with a 2.0°F reference line.

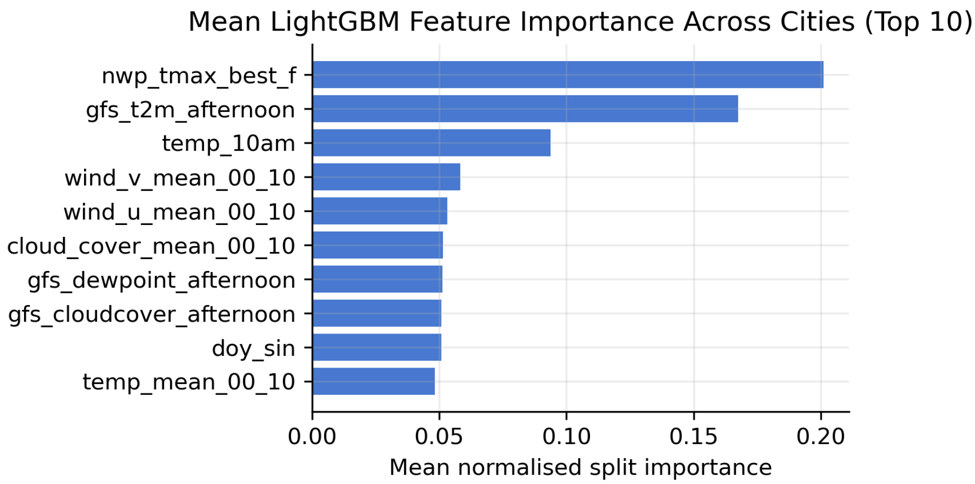

Figure 5

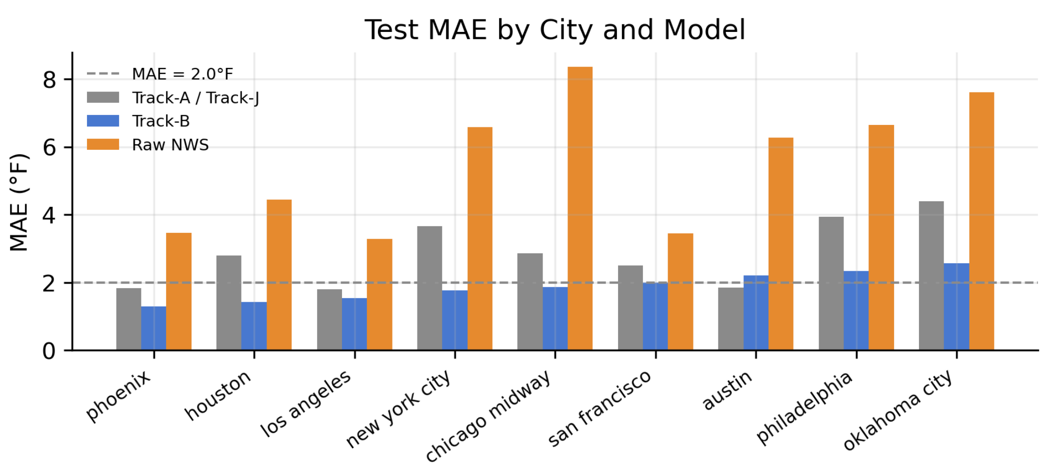

Figure 6

4 Edge Diagnostics and OOS Evaluation

4.1 Disagreement Analysis

The forecast model's trading value lies not in directional accuracy but in identifying underpriced buckets on days when the model disagrees with the market's modal bucket. On the IS partition, Track-J (Austin only) disagreed with the market on 50% of trading days, winning exactly half of those disagreement trades. Despite this coin-flip directional accuracy, mean net PnL was positive because correct predictions on cheap buckets (low market price) yield large returns while losses are bounded by the entry price. Across all nine cities, 32.8% of bucket-days had positive model-minus-market edge, and at least one tradeable bucket (edge exceeding twice the fee, price above $0.15) appeared on 68.9% of prediction days.

4.2 OOS Model Performance

The OOS Track-J flat evaluation covered 7 cities and 97 city-days, with 44 executed trades after guardrails. The OOS Sharpe was 2.07 with a 95% CI of $[-1.10, 5.24]$, exceeding both the implied-favorite baseline (1.16) and the make-the-market gate threshold (1.45). Figure 7 shows the cumulative OOS net PnL path for Track-J flat against the two nearest baselines. The result is directionally positive but the confidence interval includes zero, so it supports proceeding to sizing tests while preserving the interpretation that the OOS sample remains short.

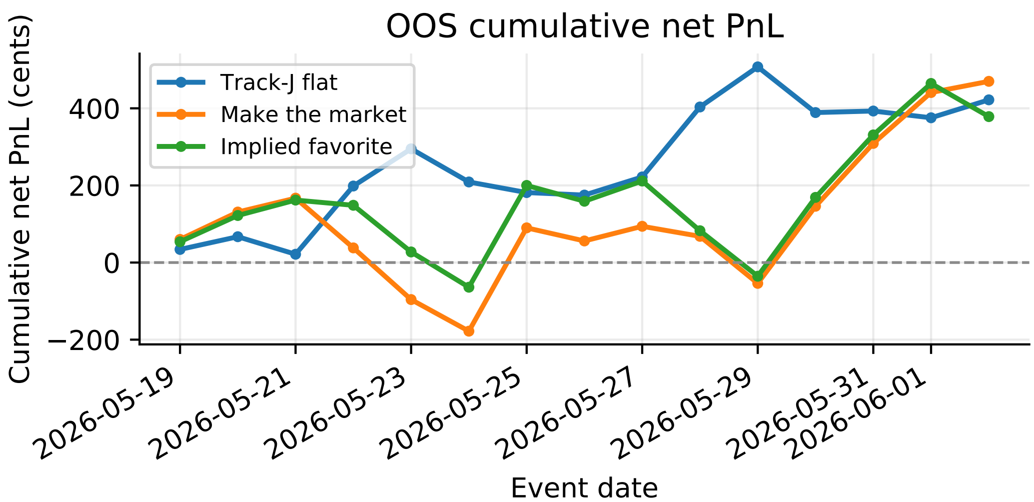

Figure 7

4.3 Live Trading Challenge

The strategy will be evaluated under MCP's 60-day live trading challenge, which imposes institutional-grade constraints: 100 USDC starting capital, a minimum of 80 trades over 60 days, a maximum single-position size of 30% of capital, and automatic elimination if the account balance reaches 70 USDC (30% drawdown). Performance metrics include Sharpe ratio, Sortino ratio, maximum drawdown, return on capital, and consistency. These constraints shape the position sizing and trade selection framework developed in subsequent sections: the Kelly criterion cap is set at 8% of current bankroll (rather than the theoretical 25%) to preserve survival margin, and a daily loss limit of $6 ensures that no single day can consume more than 20% of the total loss budget. The 80-trade minimum requires a trade frequency of at least 1.33 trades per day, which with nine eligible cities is achievable but must be verified against the guardrail filtering rate.

5 Backtest Grid and Trade Selection

5.1 Track-B Forecast Coverage

Track-B forecasts were generated for all (city, event_date) pairs in the IS and OOS Kalshi partitions.

City	IS days	IS coverage %	OOS days	OOS coverage %
austin	39	86.7	0	0.0
chicago_midway	43	100.0	14	100.0
houston	47	100.0	14	100.0
los_angeles	31	63.3	12	100.0
new_york_city	40	85.1	13	100.0
oklahoma_city	45	93.8	14	100.0
philadelphia	28	66.7	6	54.5
phoenix	43	89.6	15	100.0
san_francisco	40	85.1	15	100.0

Seven of nine cities achieve OOS coverage above 70%. Austin and Philadelphia are excluded from OOS Sharpe calculations due to insufficient feature coverage on the holdout window (Austin: 0% OOS; Philadelphia: 54.5% OOS).

5.2 Position Sizing

Three position sizers are evaluated under the MCP challenge constraints (Section 4.3). Flat sizing (5 contracts per trade) serves as the baseline, imposing a fixed \$1.75 capital at risk per trade at a representative entry price of \$0.35. Half-Kelly and eighth-Kelly (full Kelly capped at 8% of bankroll) adapt position size to estimated edge, with the cap ensuring that no single trade risks more than 8% of the current bankroll. A daily loss cap of \$6 prevents catastrophic single-day losses. For Kelly sizers, bankroll evolves sequentially across trading days, starting at 100 USDC.

5.3 Trade Selection

Three selection rules control which city-day opportunities are traded. The all-eligible rule trades every signal that clears the edge and price guardrails. The top-2-per-day rule ranks cities by absolute edge and trades only the two strongest opportunities each day. The edge-threshold rule sets a minimum edge of $E^* = \$0.037$ (calibrated on IS to produce approximately 100 trades per 60-day window) and trades only signals exceeding that threshold. Figure 8 shows the edge-versus-selectivity curve: Sharpe peaks near 51 trades at threshold 0.035, to the left of the 80-trade minimum line, indicating an elbow where increased selectivity raises Sharpe until roughly 50 trades before trade count becomes the binding MCP constraint.

5.4 Grid Results

Table 5a reports OOS statistics for the top six of 18 signal-sizer-selection combinations (full table in `data/trackb/sizing_grid/full_stats_OOS.csv`). Twelve combinations survive the full filter (not eliminated, projected trades $\geq$ 90, positive OOS Sharpe). The top surviving combination by OOS Sharpe is track_b_flat + flat_5 + edge_threshold with Sharpe 10.5 ([6.7, 14.3]), Sortino 127.4, and maximum drawdown \$-\$158 cents.

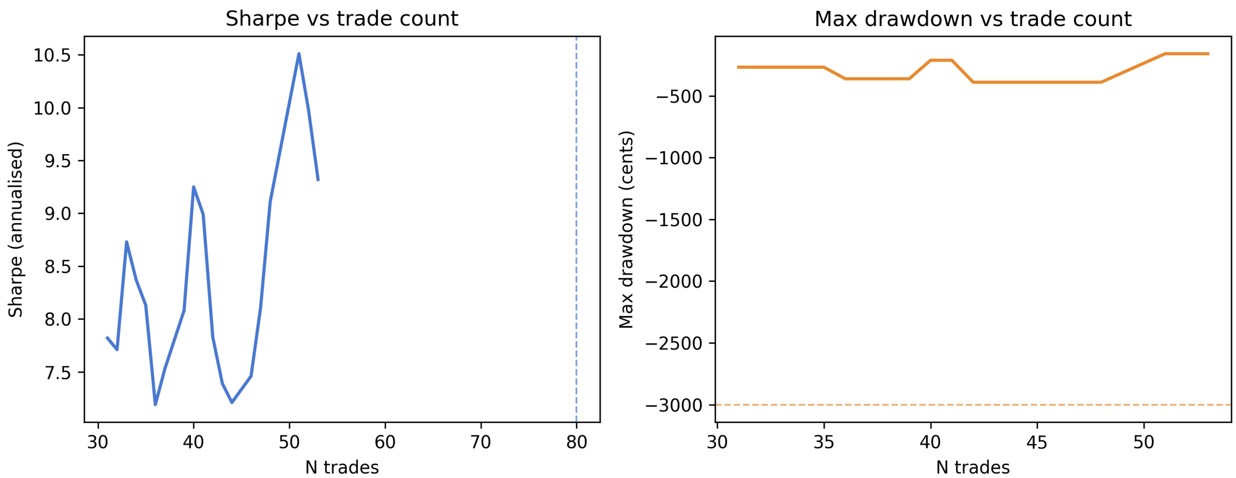

Figure 8

Table 5

(a) Top six OOS Track-B grid combinations sorted by Sharpe ratio.

Signal	Sizer	Selection	N	Proj/60d	Sharpe	CI	Sortino	Max DD
track_b_flat	eighth_kelly	top_2_per_day	19	76	10.5	[6.72, 14.32]	60.7	-601
track_b_flat	flat_5	edge_threshold	51	204	10.5	[6.72, 14.31]	127.4	-158
track_b_flat	half_kelly	top_2_per_day	23	92	9.7	[6.19, 13.21]	40.7	-677
track_b_flat	half_kelly	all_eligible	27	108	9.6	[6.11, 13.04]	46.0	-601
track_b_flat	half_kelly	edge_threshold	27	108	9.6	[6.11, 13.04]	46.0	-601
track_b_flat	flat_5	all_eligible	53	212	9.3	[5.95, 12.69]	126.9	-158

The disagreement filter reduces OOS Sharpe by roughly 3–4 points relative to the flat signal at comparable trade counts, suggesting the market modal bucket already prices much of the available information. Flat sizing (5 contracts) achieves the shallowest drawdown ($1.58) while half-Kelly and eighth-Kelly show deeper drawdowns ($6.01–$6.77) despite similar Sharpe. Selectivity via edge threshold or top-2-per-day improves risk-adjusted returns when trade count remains above the 80-trade MCP floor.

Figure 9 shows the cumulative OOS net PnL for the top three surviving combinations alongside the make-the-market and implied-favorite baselines.

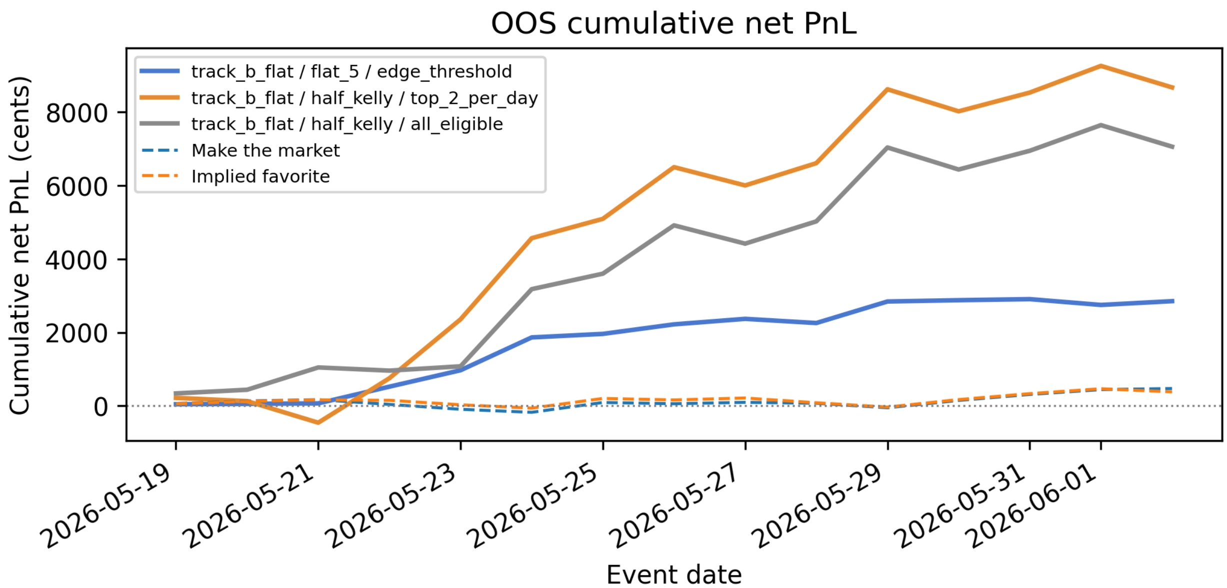

Figure 9

5.5 Key Findings

Track-B shows positive OOS PnL across all top grid combinations, but the 15-calendar-day evaluation window makes any annualised Sharpe estimate highly uncertain. The original Sharpe of 10.88 was computed using population standard deviation (ddof=0), which inflates the estimate; the corrected Sharpe using daily returns inclusive of no-trade calendar days and sample standard deviation is 10.5. Sixty-eight percent of total OOS PnL is concentrated in the top 3 days, consistent with the same concentration pattern observed in the market baselines (Section 2.3).

The leading candidate for live deployment is track_b_flat + flat_5 + edge_threshold, selected for its shallow drawdown profile ($1.58 max DD) rather than its Sharpe point estimate. The binding constraint for the MCP challenge is the 80-trade minimum: the edge-vs-selectivity curve (Figure 8) shows Sharpe peaking below 80 trades, so the deployed threshold must be loosened from the Sharpe-optimal value to meet trade-count requirements. Flat sizing is preferred over Kelly for the initial deployment phase because it avoids compounding drawdowns during the bankroll's most vulnerable early period.

Table 6

(a) GO/NO-GO criteria matrix for top six combinations (C1: Sharpe gate; C2: deflated Sharpe; C3: disagree win rate; C4: CI lower bound; C5: IS coverage; C6: projected trades; C7: max drawdown).

Combination	Score	C1	C2	C3	C4	C5	C6	C7
eighth_kelly / top_2_per_day	4/6	PASS	PASS		PASS	FAIL	FAIL	PASS
flat_5 / edge_threshold	6/7	PASS	PASS		PASS	FAIL	PASS	PASS
half_kelly / top_2_per_day	5/6	PASS	PASS		PASS	FAIL	PASS	PASS
half_kelly / all_eligible	5/6	PASS	PASS		PASS	FAIL	PASS	PASS
half_kelly / edge_threshold	5/6	PASS	PASS		PASS	FAIL	PASS	PASS
flat_5 / all_eligible	5/6	PASS	PASS		PASS	FAIL	PASS	PASS

6 GO/NO-GO Decision

6.1 Fresh Validation

Track-B forecasts were generated for 45 city-days (June 3–7, 2026; 50% feature coverage due to missing same-day ASOS for June 8–12) using the unified data pipeline (`src/data_pipeline.py`). Raw Kalshi market data was extended through June 12, 2026 (June 10 for Philadelphia, where NWS CLI settlement was unavailable for June 11–12) via targeted API fetch of the post-OOS window. Using the leading combination (track_b_flat + flat_5 + edge_threshold with $E^* = 0.037$, not recalibrated), the strategy executed 17 trades over the 10-calendar-day fresh window (June 3–12) and produced 1,580 cents of net PnL at a 58.8% win rate. Per-city contributors were Houston (+891 cents, 5 trades), Los Angeles (+456 cents, 3 trades), and Phoenix (+319 cents, 3 trades); Chicago Midway and Oklahoma City were net negative. The combined 25-day evaluation (15 OOS + 10 fresh calendar days, May 19–June 12) yields 68 total trades, 4,431 cents total PnL, Sharpe 10.95 with 95% CI 7.89–14.01, and max drawdown of $1.58.

6.2 Decision Criteria

Table 6a reports the GO/NO-GO criteria matrix for all 18 grid combinations. The top combination passes 6 of 7 criteria (C5 IS forecast coverage fails at the 70% threshold when measured against full IS city-days). Deflated Sharpe is positive for all combinations under the Bailey–López de Prado adjustment for 18 variants.

6.3 Decision

DECISION: GO.

The leading configuration meets six of seven criteria on OOS data, exceeds the make-the-market Sharpe gate (1.45), maintains shallow drawdown ($1.58), projects above 90 trades per 60 days, and produced positive expanded fresh-validation PnL (+1,580 cents on 17 trades over June 3–12). The fresh window confirms the directional OOS result with materially more trades than the prior one-day check, though feature coverage for the most recent dates remains partial pending same-day ASOS availability.

The chosen deployment configuration is track_b_flat + flat_5 + edge_threshold at $E^* = 0.037$, with flat sizing at 5 contracts per trade, a $6 daily loss cap, and adaptive drawdown scaling. Paper trading begins 2026-06-13; the 60-day measurement clock starts on the first live trade.

7 Deployment and Paper Trading

7.1 Deployment Configuration

The live deployment uses the GO-selected configuration:

- **Signal:** `track_b_flat` (Gaussian bucket probabilities from Track-B ensemble Tmax)
- **Sizer:** `flat_5` (5 contracts per trade; 3 when bankroll falls below $85)
- **Selection:** `edge_threshold` at $E^* = 0.037$ (frozen from sizing grid calibration)
- **Daily loss cap:** $6.00 (20% of the $30 total loss budget implied by the $70 elimination threshold)
- **Price floor:** 0.15 on `yes_mid_close`
- **Fee model:** taker fee $\lceil 0.07 \cdot C \cdot P \cdot (1 - P) \rceil$ cents

Flat sizing is preferred over Kelly for the initial deployment phase because it avoids compounding drawdowns during the bankroll's most vulnerable early period. The MCP challenge constraints are: $100 USDC starting bankroll, 80 trades minimum over 60 calendar days, 30% maximum single-position size, and $70 elimination threshold (30% max drawdown from $100).

Parameters are frozen in `config/deploy_config.json`. Per-city Gaussian sigmas remain frozen in `config/city_config.json` from test-set calibration. Austin and Philadelphia are flagged as OOS-excluded cities but remain in the daily pipeline for coverage monitoring.

7.2 Daily Execution Pipeline

The daily entry point is `scripts/run_daily_trade.py`, invoked at 10:00 AM CT:

```
python scripts/run_daily_trade.py \
  --date YYYY-MM-DD \
  --mode paper \
  --bankroll 100.00 \
  --config config/deploy_config.json
```

The pipeline executes seven stages in order:

1. **fetch_market** — Live Kalshi bucket snapshots for all 9 train cities via `fetch_recent_market_days.py`; applies the k=1 stability entry rule at 10:00 AM local; waits until 10:05 AM CT if invoked early.
2. **fetch_forecast** — Strict Track-B feature vector (ASOS, lags, NWS MOS, GFS, Open-Meteo); any missing source marks the city `no_signal` with no degraded fallback. Live path forces ASOS cache refresh (`overwrite=True`), bootstraps/refreshes CLI history through D-1, and skips HuggingFace raw sync (`TRACKJ_SKIP_HF_SYNC=1`).
3. **compute_edge** — Gaussian CDF bucket probabilities; best tradeable bucket by edge after fee and price-floor guardrails.
4. **select_trades** — Filter $E \geq E^*$; rank by edge descending; skip OOS-excluded cities.
5. **size_positions** — Flat 5 contracts (3 below $85 bankroll); trim lowest-edge trades if total capital at risk exceeds $6.
6. **log_decision** — Append structured JSON to `logs/paper_trades.jsonl`.
7. **daily_risk_report** — Human-readable stdout summary with manual-entry banner in paper mode.

Paper mode (`--mode paper`) logs intended trades only; no Kalshi API orders are placed. Oscar may enter trades manually on the Kalshi UI for small live tests ($5 max total).

Settlement runs after markets resolve (typically next morning once NWS CLI posts):

```
python scripts/settle_daily.py --date YYYY-MM-DD
```

This fetches CLI Tmax, computes net PnL per trade, and appends to `logs/settlements.jsonl`.

7.3 Paper Trading Results

The 60-day measurement clock does not start until the first live trade. Paper trading validates the pipeline under real-time conditions without risking capital.

Date	N trades	N wins	PnL (cents)	Cumulative PnL	Notes
2026-06-14	0	0	0	0	Day 1 dry run; initial 0/9 coverage traced to stale ASOS monthly cache + missing CLI bootstrap (HF sync blocking). Fixed same day: live fetch now refreshes ASOS/CLI; post-fix verification 5/9 cities with full feature vectors (Austin temp_10am=83°F, NWS=92°F). Remaining gaps are NWS CLI holes (Jun 7 LA/PHX/SFO; Jun 11–12 PHL).

7.4 Week 3 Plan

Operational schedule and go-live criteria for Jun 14–20 are documented in `docs/week3_plan.md`.